

YUZHOU (EMRYS)

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PROFILE

Quantitative researcher and data analyst with advanced training in applied econometrics, large-scale microdata analysis, and causal inference. PhD candidate in Economics at ANU, with a peer-reviewed publication in a Q1 journal based on large-scale administrative student-level data. Proficient in Python and R across the full data pipeline — data acquisition, cleaning, reshaping, modelling, and visualisation. Strong written and oral communication skills developed through academic publishing, undergraduate teaching, and professional committee work. Background spans economics, social science, and education policy.

TECHNICAL SKILLS

Software & Programming Languages

Python: pandas (data loading, merging, groupby aggregation, reshaping, missing data handling), numpy (array operations, statistical computations), matplotlib and seaborn (publication-quality charts, distribution plots, heatmaps), scipy (hypothesis testing: t-tests, chi-square, ANOVA), statsmodels (OLS/WLS regression, robust standard errors, model diagnostics); scripting for reproducible end-to-end data workflows

R: tidyverse suite: dplyr (data manipulation, filtering, summarising, joins), tidyr (reshaping between wide/long formats, nesting), ggplot2 (layered grammar-of-graphics visualisations, multi-panel figures), readr/readxl (data ingestion from multiple file formats); fixest (high-dimensional fixed effects regression, clustered standard errors), sandwich and lmtest (heteroskedasticity-robust and cluster-robust inference); R Markdown and knitr for reproducible analytical reports

Excel: advanced modelling, pivot tables, data validation, formula-driven reporting

Statistical & Analytical Methods

- Causal inference: instrumental variables (IV), regression discontinuity design (RDD), difference-in-differences (DiD) — applied to large-scale administrative panel datasets in Python and R
- Regression modelling: OLS, WLS, fixed/random effects, clustered and heteroskedasticity-robust standard errors; model diagnostics and sensitivity analysis
- Large-scale microdata management: merging and reshaping multi-source administrative datasets, deduplication, missing data handling, outlier detection, and validation in Python (pandas) and R (dplyr/tidyr)
- Data visualisation and reporting: publication-quality figures (ggplot2, seaborn, matplotlib); reproducible analytical reports (R Markdown, knitr)

EDUCATION

2022 – Present PhD in Economics — Australian National University, Canberra

Applied microeconomics; labour economics; causal inference with large-scale administrative data.

Expected submission: late 2026.

2020 – 2022 Master of Economics — Australian National University, Canberra

Econometric methods, statistical modelling, economic policy analysis.

2016 – 2020 Bachelor of Auditing — Shanghai Lixin University of Accounting and Finance

2021 Graduate Certificate of Finance — Antwerp School of Management

2020 Securities Analyst Qualification — China Securities Association

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Researcher & Economics Tutor | Australian National University, Canberra | 2022 – Present

- Constructed and managed a large-scale administrative panel dataset (multi-cohort, student-level records) in Python and R; performed data cleaning, merging across multiple source files, deduplication, and repeated validation checks to ensure analytical integrity.
- Applied causal inference methods to address endogeneity: IV estimation and regression discontinuity design (RDD) exploiting discrete academic ranking thresholds to isolate causal effects of peer achievement on tutoring investment.
- Ran the complete analytical pipeline in Python (pandas, statsmodels) and R (fixest, ggplot2) — from raw administrative data through econometric modelling, robustness checks, and output automation — delivering publication-ready tables and figures.
- Produced a peer-reviewed publication in a Q1 journal (Economics of Education Review, 2025) translating complex statistical findings into evidence-based policy conclusions on educational inequality and shadow education.
- Tutored ~20 undergraduate Economics students per semester; designed assessments, marked work, and provided structured written feedback to improve quantitative understanding.

Ordinary Committee Member | Economic Society of Australia (ACT Branch) | 2025 – Present

- Plan and coordinate professional seminars connecting economists, policymakers, and researchers across government, academia, and the private sector.
- Manage stakeholder correspondence, meeting logistics, and event promotion.

Academic Mentor — Toad Hall | Australian National University | 2024 – Present

- Provide structured one-on-one mentoring and written feedback to undergraduate residents; adapt approach to individual learning needs and academic circumstances.

Analyst Intern — Smart Education Department | PingAn Technology Shenzhen Co., Ltd. | Dec 2019 – Feb 2021

- Maintained and quality-checked administrative datasets for education product analytics within a technology-driven education platform.
- Supported strategic analysis and revenue pipeline reporting through structured data outputs and summary documentation.

Fixed Income Division Intern | Guosen Securities Co., Ltd. (China) | Jun – Aug 2021

- Reviewed bond prospectuses for 8 projects and verified 17 documentation packages for financial accuracy; all cleared without material error.

Commercial Insurance Intern | AXA Tianping Co., Ltd. (China) | Aug – Nov 2019

- Managed commercial budget accountabilities and policy issuance; monitored campaign results across 20+ accounts.

PUBLICATION

Dai, C., & Zhou, Y. (2025). Rank, peer achievement, and shadow education: Evidence from secondary school students in China. *Economics of Education Review*, Vol. 108, Oct. 2025, 102682.

ACHIEVEMENTS & LANGUAGES

2023–2027 Postgraduate Research Scholarship, Australian National University

2023–2027 HDR Fee Remission Merit Scholarship, Australian National University

2019 National English Competition for College Students — First Prize

2018 4th National College Student Financial Elite Challenge — Third Prize

2017–2020 Multiple Academic Scholarships (Undergraduate)

Languages: Advanced English (IELTS 7.5, professional fluency); Mandarin (native speaker)

REFEREES

Available on request.